

Advances in High-Spectral-Efficiency Formats: Modulation and Coding

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(Invited Paper)

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Abstract Probabilistic amplitude shaping, increasingly adopted in coherent systems, relies on bit-metric decoding that introduces residual demapping loss. This paper shows that single parity-check coded probabilistic shaping reduces this loss via iterative demapping, achieving up to 0.9 dB SNR gain experimentally confirmed in long-haul WDM transmission. ©2026 The Author(s)

Introduction

Capacity scaling in coherent optical communication relies on a combination of higher symbol rates and improved spectral efficiency (SE). Advances in high-speed electronic integration have enabled single-carrier operation beyond 100 GBd [1–3], reducing the number of wavelengths required to achieve a given throughput and, consequently, lowering cost per bit, particularly in short- and metro-reach systems. At the same time, higher symbol rates place increasing demands on the analog front end and associated digital signal processing, including bandwidth limitations, finite effective number of bits in high-speed data converters, and power consumption [4, 5].

Once a symbol-rate operating point is selected based on front-end or integration constraints, increases in throughput require higher SE [6].

Probabilistic amplitude shaping (PAS) [7] improves SE by tailoring the transmitted amplitude distribution to better match the Gaussian channel while maintaining compatibility with QAM constellations and binary forward-error-correction (FEC) codes. In practical systems, PAS is combined with bit-metric decoding (BMD), which operates on bitwise reliability information rather than symbol-level metrics and introduces a performance gap relative to the achievable information rate under symbol-metric decoding (SMD), referred to here as residual demapping inefficiency.

Importantly, the performance impact associated with this gap persists largely independent of the selected symbol rate. As a result, even partial mitigation under fixed operating assumptions can be interpreted as increased tolerance to noise or equivalently as operation at higher SE for a given signal-to-noise ratio (SNR).

This paper examines residual demapping inefficiency in high-SE probabilistically shaped signaling. Single parity-check (SPC)-coded probabilistic shaping [8, 9] is investigated as a controlled means of introducing structured redundancy at the modulation interface to enable iterative demapping and decoding under fixed operating assumptions.

Fixed System Model for Analysis

All evaluations are conducted under a fixed system model to ensure consistent interpretation. Symbol rate, modulation order, and FEC parameters are held constant across all cases so that observed performance differences are attributed solely to changes in the coded-modulation structure rather than to symbol-rate or FEC variations.

At the receiver, the analysis focuses on soft-decision demapping and decoding. Iterative information exchange between the demapper and the LDPC decoder is evaluated using fixed LDPC parameters to isolate its impact on achievable SE.

SPC-Coded Probabilistic Shaping

SPC-coded probabilistic shaping extends conventional PAS-based QAM by introducing structured redundancy on selected bit positions while preserving the target amplitude statistics. The standard PAS framework and LDPC code are retained, with modifications limited to the shaping–mapping interface.

Figure 1 shows the transmitter and receiver architecture used for SPC-coded probabilistic shaping, highlighting the application of SPC constraints on PAS sign bits and the resulting iterative demapper–decoder exchange.

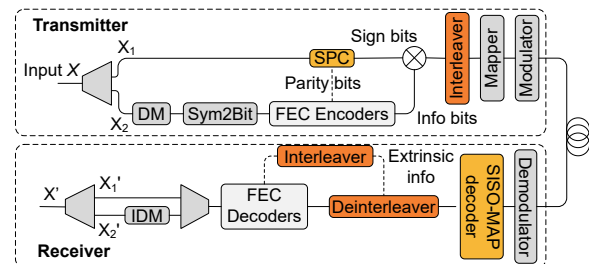


Fig. 1: Block diagram of the SPC-coded probabilistic shaping transmitter and receiver.

In PAS, symbol amplitudes are drawn from a non-uniform distribution generated by a distribution matcher (DM), while sign bits remain uniformly distributed. This separation provides a natural interface for introducing additional coded structure without modifying the amplitude statistics. In the

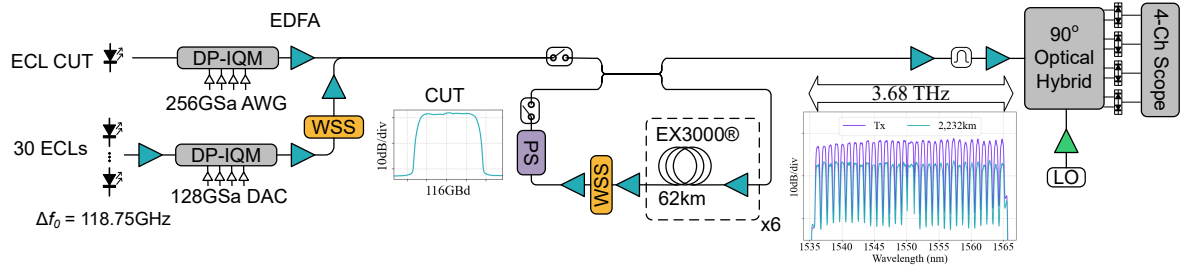


Fig. 2: Experimental setup for 3.68-THz WDM transmission. Left inset: optical spectrum of the CUT at the transmitter. Right inset: full WDM spectrum at the transmitter and after 2232 km of transmission.

SPC-coded approach, short SPC codes with a single parity constraint per sign-bit group (e.g., SPC(3,4) or SPC(7,8)) are applied prior to symbol mapping. Because sign bits are equiprobable, and parity bits of linear block codes are themselves uniformly distributed [10], the imposed SPC parity constraints preserve the output distribution of the DM and the target constellation shape.

The small rate loss introduced by the SPC constraint is compensated by adjusting the shaping factor so that the net SE, modulation alphabet, and LDPC code remain unchanged relative to the baseline PAS case.

At the receiver, a soft-input/soft-output MAP decoder incorporates the SPC constraint when computing bit log-likelihood ratios (LLRs) and exchanges extrinsic information with the LDPC decoder over a small number of outer iterations. The LDPC code is identical across all cases so that observed performance differences reflect only changes in the coded-modulation structure.

System Performance

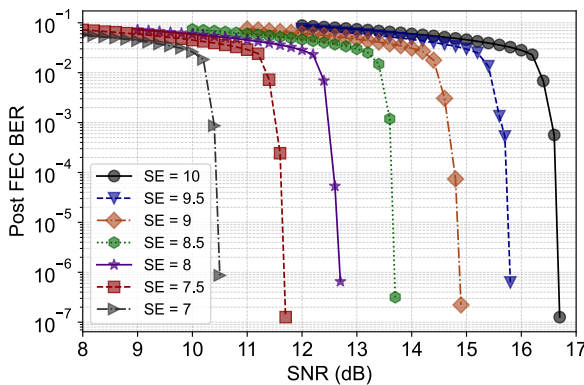


Fig. 3: Simulation results: Post-FEC BER versus SNR for SPC-coded PS-256QAM at SEs from 7 to 10 b/s/Hz.

Performance of SPC-coded probabilistic shaping was evaluated through numerical simulations and long-haul WDM transmission experiments. All results are assessed relative to a baseline PAS receiver employing BMD, under identical modulation order, FEC code, coding overhead, and symbol rate.

Simulation results in Fig. 3 show the performance of SPC-coded PS-256QAM after four outer demapping iterations and ten inner LDPC itera-

tions. Compared to conventional PS-256QAM operating at the same SE and using a total of 40 LDPC iterations (Fig. 4), the SPC-coded scheme consistently requires lower SNR to achieve a given target SE. Across the evaluated operating range, an SNR reduction of up to approximately 0.9 dB is observed relative to the baseline PAS receiver, with the maximum gain occurring at 9 b/s/Hz. This improvement is attributed to the iterative exchange of extrinsic information between the demapper and the FEC decoder enabled by the SPC constraint.

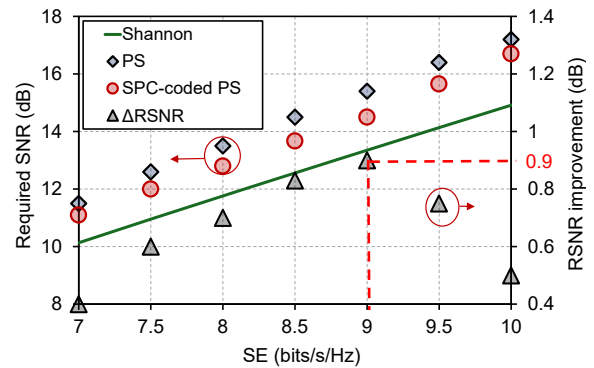


Fig. 4: Required SNR vs SE for PS-256QAM and SPC-coded PS-256QAM.

As shown in [8], extensive experimental data confirm that error-correction thresholds closely align with simulations, within 0.10 dB to 0.15 dB, providing prior experimental validation of SPC-coded probabilistic shaping.

Experimental validation was carried out in a long-haul WDM system operating at 116 GBd using a recirculating-loop configuration, as shown in Fig. 2. The system employed 31 dual-polarization WDM channels spaced at 118.75 GHz, covering a total bandwidth of 3.68 THz, with one channel designated as the channel under test (CUT) and swept across the spectrum. Transmission was performed over standard single-mode fiber using six 62-km spans per loop, with C-band EDFAs maintaining a total launch power of 19 dBm and wavelength-selective switches providing power equalization and gain-tilt compensation.

Long-haul transmission results, shown in Fig. 5, confirm that the gains predicted by simulation are preserved under realistic WDM propagation conditions. After 1860 km, corresponding to

five recirculating-loop passes, SPC-coded PAS achieved an average net SE of 7.5 b/s/Hz, yielding an aggregate throughput of 26.97 Tbps. At 2232 km (six loop passes), the net SE decreased slightly to 7.31 b/s/Hz, resulting in an aggregate throughput of 26.29 Tbps. These results confirm that the iterative demapping gains observed in simulation persist beyond back-to-back operation and remain effective in the presence of fiber nonlinearities, amplifier noise, and inter-channel interference.

Figure 5 presents the estimated SNR and computed NGMI after 1860 km (hollow markers) and 2232 km (solid markers), respectively. The top panel shows the estimated SNR as a function of wavelength, with marker colors indicating the successfully received SEs: green for 8 b/s/Hz, purple for 7.5 b/s/Hz, and teal for 7 b/s/Hz. Across the occupied bandwidth, the SNR variation remains modest, enabling consistent support of the targeted SEs.

The bottom panel shows the corresponding NGMI values, which follow a similar wavelength-dependent trend. NGMI remains consistently above the FEC decoding threshold for all tested SEs at both transmission distances, confirming reliable transmission performance and indicating that the iterative demapping gains observed in simulation translate into robust operation in long-haul WDM propagation.

Relative to PAS under identical transmission conditions, the reduced SNR requirement translates into additional operating margin in long-haul transmission. Experimentally, this margin can be traded for increased transmission reach or higher net SE under fixed operating conditions. These results demonstrate that SPC-coded probabilistic shaping delivers tangible system-level gains while preserving compatibility with existing PAS-based transceiver architectures.

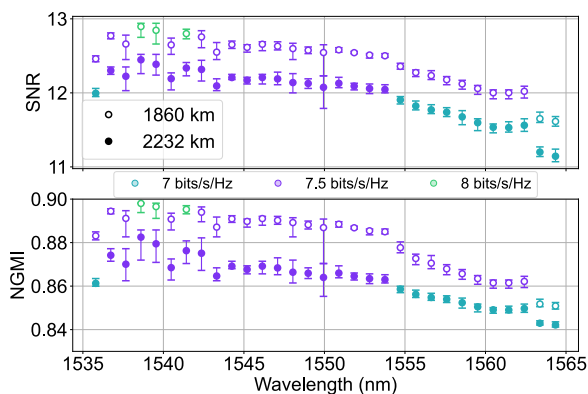


Fig. 5: Estimated SNR and NGMI after 1860 km and 2232 km. Error bars show measurement variability.

Implications for Capacity Scaling

Under fixed symbol-rate and FEC constraints, SPC-coded probabilistic shaping consistently out-

performs conventional PAS, providing measurable SNR gains across a range of spectral efficiencies. These gains are achieved without modifying the modulation order, FEC code, or analog front end, and therefore preserve the underlying transceiver operating point.

The results presented here show that introducing controlled structure at the modulation interface provides a practical means to improve performance under otherwise identical system assumptions. In the demonstrated long-haul WDM system, the reduced SNR requirement translates directly into increased operating margin, which can be traded for extended transmission reach or higher net SE under fixed launch power and bandwidth conditions.

From a system perspective, this approach complements symbol-rate scaling rather than replacing it. While increases in symbol rate remain a primary driver of capacity growth, their effectiveness is increasingly constrained by power consumption and implementation complexity. Improving coded-modulation efficiency through iterative demapping enabled by SPC-coded probabilistic shaping offers an additional, largely orthogonal path to throughput scaling, particularly in regimes where further symbol-rate increases yield diminishing returns.

Conclusion

This work investigated the performance benefits of SPC-coded probabilistic shaping in high-spectral-efficiency coherent transmission under fixed symbol-rate and FEC constraints. By introducing short SPC constraints on PAS sign bits, iterative demapping and decoding are enabled without altering the target amplitude distribution, modulation order, or LDPC code.

Numerical simulations and long-haul WDM transmission experiments show that SPC-coded probabilistic shaping reduces the SNR required to achieve a given net SE by up to 0.9 dB relative to conventional PAS with bit-metric decoding. While PS-256QAM is used in this work due to its relevance for high-spectral-efficiency operation, the proposed SPC-coded shaping framework is not specific to the modulation order. These gains persist under realistic transmission impairments and translate directly into increased operating margin, extended reach, or higher throughput under otherwise identical system conditions.

More broadly, the results demonstrate that improving the coded-modulation interface through structured redundancy and iterative demapping enhances the efficiency of PAS-based coherent systems. Such approaches complement ongoing symbol-rate scaling and front-end advances, providing an additional degree of freedom for capacity growth in future optical communication systems.

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